ClassMate

Group 11

Our Team

Front-end

Lead UI/UX Architect
Penny Ahlstrom

Lead Interaction & State Management Engineer
Nezihe Tekin

Back-end

Lead Backend Engineer
James Jones

Lead Backend Architect
Omar McIntosh

Team 11. (2025). *Requirements Analysis Design* [Internal project report]. George Brown College.

Introduction

- **Helps students stay organized** by managing courses, tasks, reminders, and grades in one centralized web app.
- **Improves academic planning** with clear views of upcoming deadlines and performance summaries through student-entered data.
- **Built on a scalable microservices foundation**, enabling future expansion into advanced features such as analytics, integrations, and enhanced productivity tools.

Team 11. (2025). *Requirements Analysis Design* [Internal project report]. George Brown College.

Backend Infrastructure & Security Architecture

API and Microservices

- REST API
- DTO
- Spring Boot
- Java 21

classmate-backend_v3 main TaskIntegrationTest

Project

- main
  - java
    - com.classmate.userservice
      - config
      - controller
      - dto
      - exception
      - model
      - repository
      - service
        - AuthService
        - JwtService
      - UserServiceApplication
    - resources
  - test
- .gitattributes
- .gitignore
- build.gradle
- Dockerfile
- gradlew
- gradlew.bat
- HELP.md
- settings.gradle
- .env
- .gitattributes
- .gitignore
- architecture.md
- build.gradle
- docker-compose.yml

e.yml | application-docker.properties | JwtService.java | AuthService.java | CourseController.java | bui

```gradle
plugins {
    id 'org.springframework.boot' version '3.5.3' // upgrade from 3.3.5
    id 'io.spring.dependency-management' version '1.1.7'
    id 'java'
}

group = 'com.classmate'
version = '0.0.1-SNAPSHOT'
description = 'user-service'

java {
    toolchain { JavaToolchainSpec it ->
        languageVersion = JavaLanguageVersion.of(21)
    }
}

configurations {
    compileOnly {
        extendsFrom annotationProcessor
    }
}

repositories {
    mavenCentral()
}

dependencies { Add Starters...

    //PENNY - Flyway
    implementation 'org.flywaydb:flyway-core'
    implementation 'org.flywaydb:flyway-database-postgresql'
```

Activate Windows
Go to Settings to activate Windows.

classmate-backend_v3 > user-service > build.gradle 1:1 CRLF UTF-8 4 spaces

Containerization and Profile

- Docker Compose
- Scalable
- Secure

classmate-backend_v3 main TaskIntegrationTest

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e.yml | application-docker.properties | JwtService.java | AuthService.java | CourseController.java | buil

```
12  public class AuthService {
18      public AuthService(UserRepository userRepository,   James Jones +1
24      }
25
26      public AuthResponseDTO register(RegisterRequestDTO request) {   1 usage   James Jones +1
27
28          if (userRepository.existsByEmail(request.email())) {
29              throw new RuntimeException("Email already in use");
30          }
31
32          User user = new User();
33          user.setFirstName(request.firstName());
34          user.setLastName(request.lastName());
35          user.setEmail(request.email());
36          user.setPassword(passwordEncoder.encode(request.password()));
37
38          User saved = userRepository.save(user);
39
40          String token = jwtService.generateToken(saved.getEmail());
41          return new AuthResponseDTO(saved.getUserId(), token);
42      }
43
44      public AuthResponseDTO login(LoginRequestDTO request) {   1 usage   James Jones +1
45
46          User user = userRepository.findByEmail(request.email())
47                  .orElseThrow(() -> new RuntimeException("invalid credentials"));
48
49          if (!passwordEncoder.matches(request.password(), user.getPassword())) {
50              throw new RuntimeException("invalid credentials");
51          }
52
53          String token = jwtService.generateToken(user.getEmail());
```

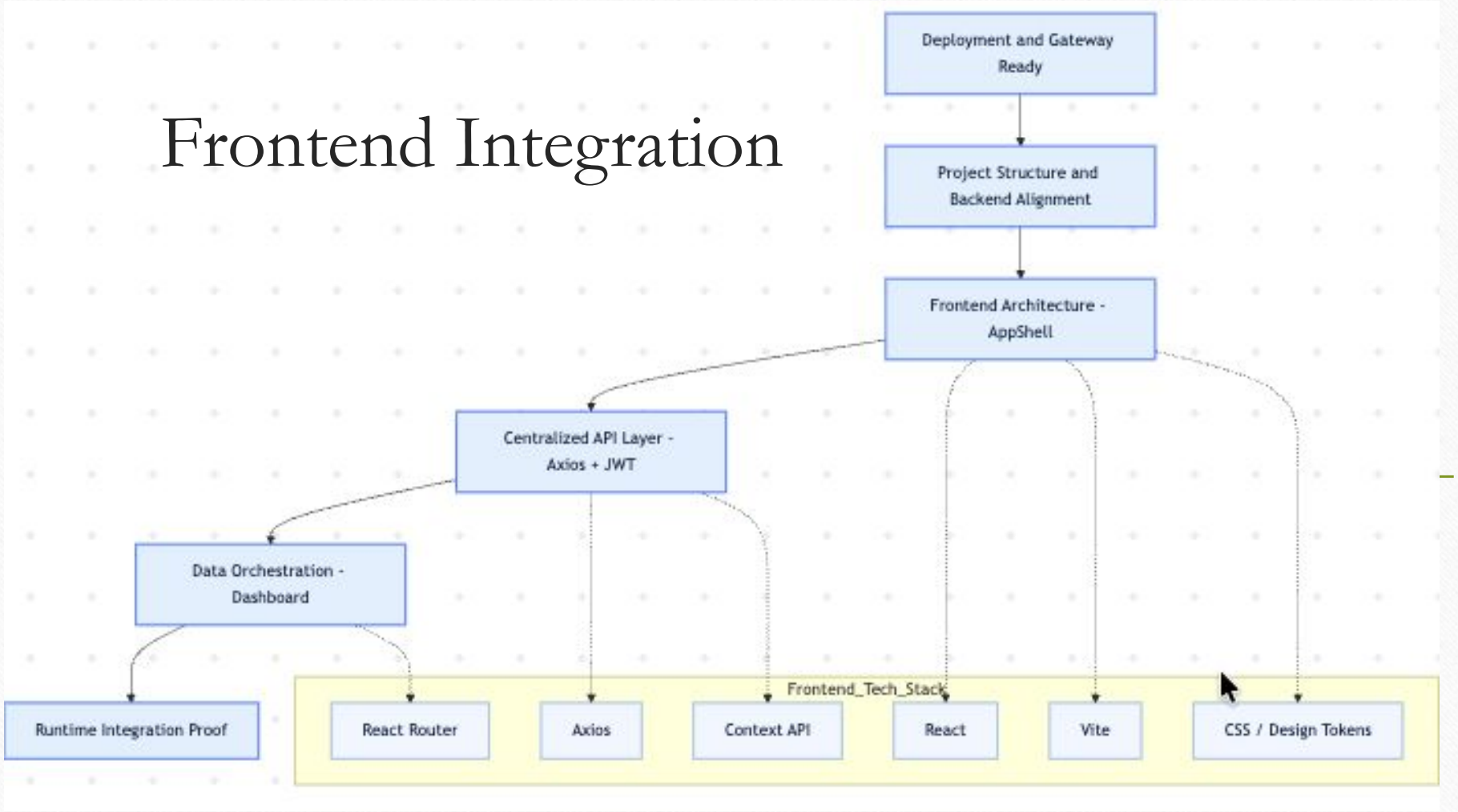
Activate Windows
Go to Settings to activate Windows.

classmate-backend_v3 > user-service > src > main > java > com > classmate > userservice > service > AuthService > register

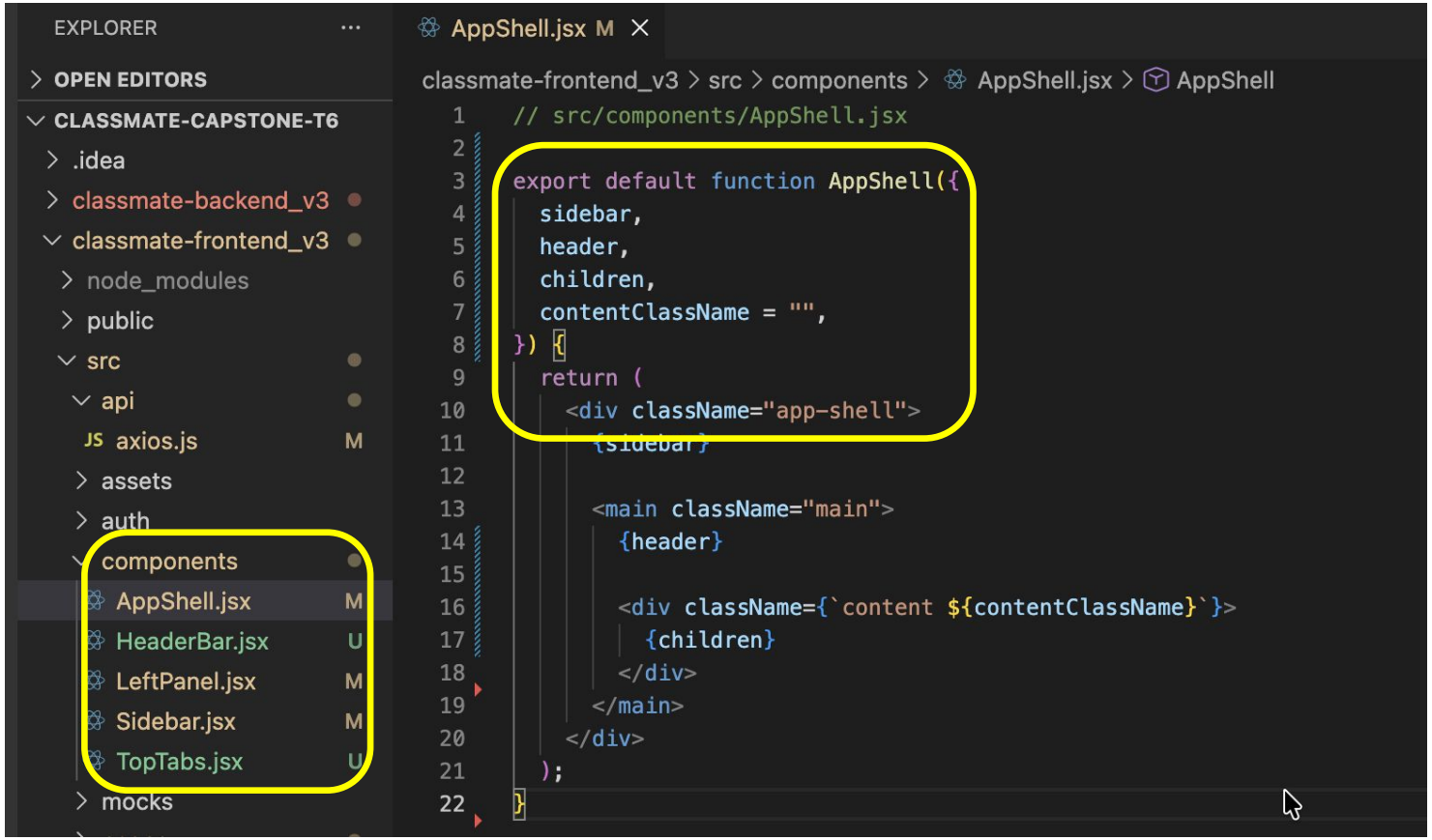
32:32 CRLF UTF-8 4 spaces

Frontend Architecture & Integration Strategy

Frontend Integration



UI Architecture & AppShell Structure



Secure API Layer & Centralized Communication

Centralized API client with automatic JWT lifecycle management

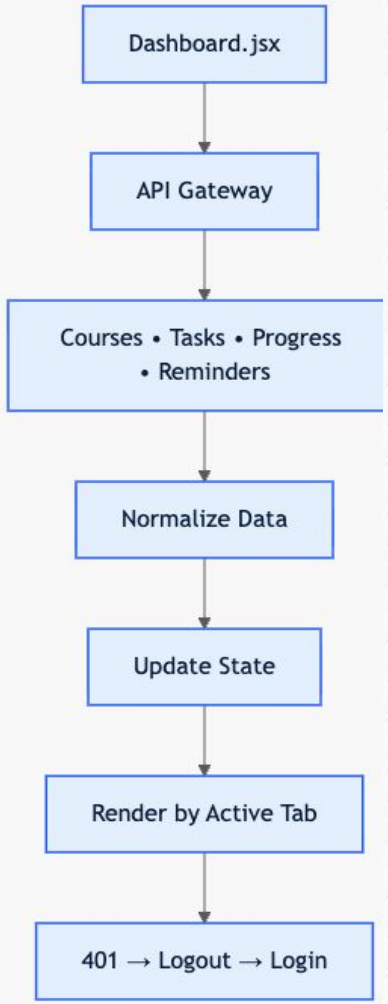
```
src
  api
    JS axios.js   M
  assets
```

```
9
10
11  const baseURL = import.meta.env.VITE_API_GATEWAY_URL || "http://localhost:8091";
12
```

JWT Attach Logic

```
19    if (url.startsWith("/api/v1/auth/")) return config;
20
21    const token = localStorage.getItem("token");
22    if (token) {
23      config.headers = config.headers ?? {};
24      config.headers.Authorization = `Bearer ${token}`;
25    }
26    return config;
27  };
```

Data Orchestration & Dashboard Implementation



Dashboard.jsx M

classmate-frontend_v3 > src > pages > Dashboard.jsx > Dashboard > createTa

```
123  export default function Dashboard() {
224    // Create flows are preserved; we only unwrap/normalize to preven
225    const createCourse = async (courseData) => {
226      const res = await api.post("/api/v1/courses", courseData);
227      const entity = normalizeCourse(unwrapEntity(res.data));
228      setCourses((prev) => [...prev, entity]);
229    };
230
231    const createTask = async (taskData) => {
232      const res = await api.post("/api/v1/tasks", taskData);
233      const entity = normalizeTask(unwrapEntity(res.data));
234      setTasks((prev) => [...prev, entity]);
235    };
236
237    const createCourseProgress = async (payload) => {
238      const res = await api.post("/api/v1/course-progress", payload);
239      const entity = normalizeProgress(unwrapEntity(res.data));
240      setProgressSnapshots((prev) => [...prev, entity]);
241    };
242
243    const createReminder = async (payload) => {
244      const res = await api.post("/api/v1/reminders", payload);
245      const entity = normalizeReminder(unwrapEntity(res.data));
246      setReminders((prev) => [...prev, entity]);
247    };
248
```

```
197          safeSet(setCourses)(c);
198          safeSet(setTasks)(t);
199          safeSet(setProgressSnapshots)(p);
200          safeSet(setReminders)(r);
201        } catch (err) {
202          if (err?.response?.status === 401) {
203            logout();
204            navigate("/login");
205            return;
206          }
```

http://localhost:5173/dashboard

repo enezihe | AWS-skill-builder... | Microsoft Azure 0... | Devops 2023 | Jobs | Active Learning in... | Data_Visualization | GeorgeBrown | Homepage - Spirit...

Dimen... 412 x 914 N 'Save

Elements | Console | Sources | Network | Performance | Memory | Application | Privacy and security | Lighthouse | Components

Preserve log | Disable cache | No throttling

Filter | Invert | More filters | All | Fetch/XHR | Doc | CSS | JS | Font | Img | Media | Manifest | Socket

50 ms | 100 ms | 150 ms | 200 ms | 250 ms | 300 ms | 350 ms

ClassMate™

Courses | Tasks

All Courses

Add a Course

Edit a Course

Delete a Course

Courses

COMP3095

Capstone Proj

UPCOMING

-1

PROGRESS

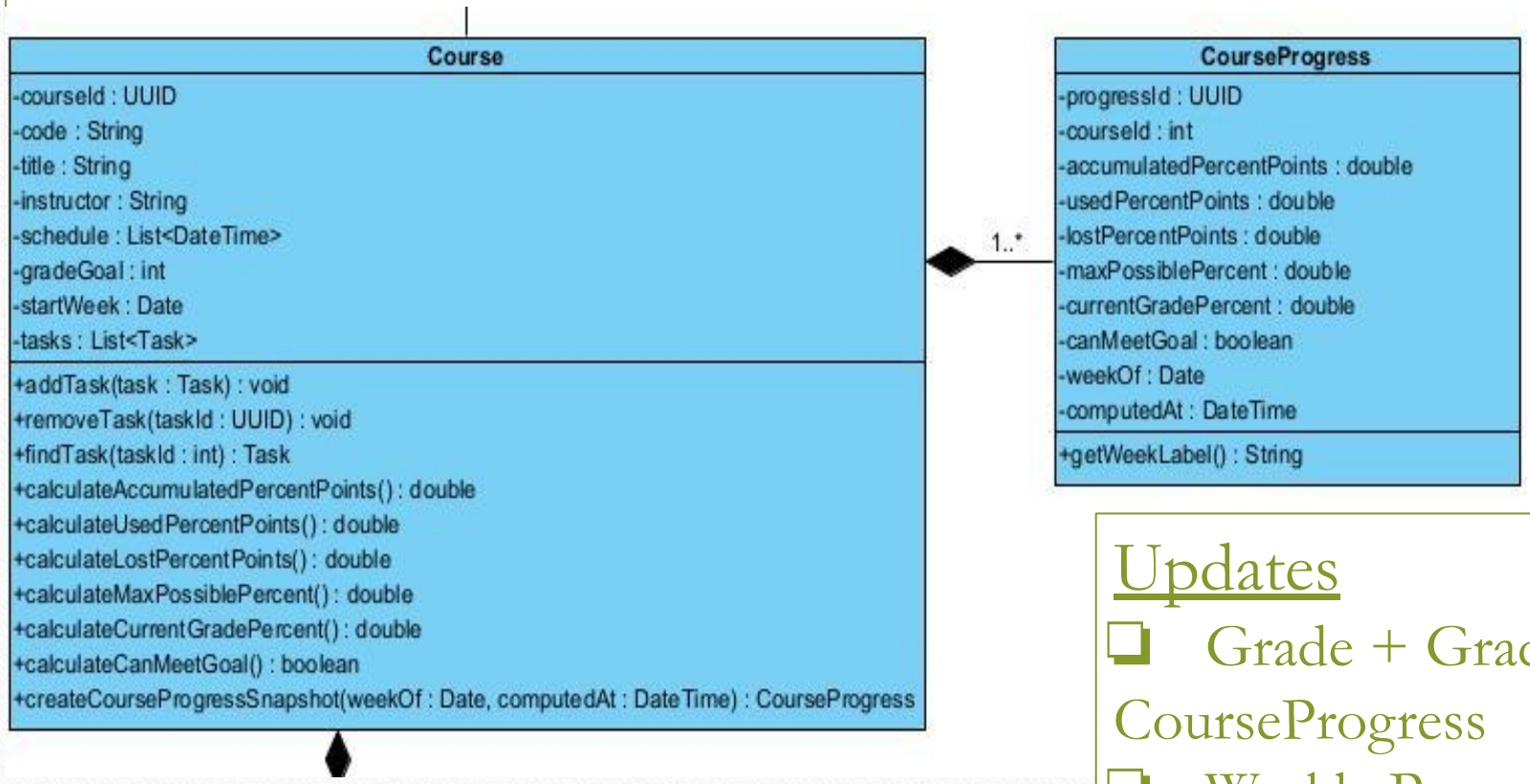
64%

+Add

Name	Status	Type	Initiator	Size	Tin
login	200	xhr	Login.jsx:21	0.7 kB	
courses	200	xhr	Dashboard.jsx:148	1.0 kB	
tasks	200	xhr	Dashboard.jsx:158	1.0 kB	
course-progress	200	xhr	Dashboard.jsx:168	3.9 kB	
reminders	200	xhr	Dashboard.jsx:178	0.8 kB	
courses	200	xhr	Dashboard.jsx:148	1.0 kB	
tasks	200	xhr	Dashboard.jsx:158	1.0 kB	
course-progress	200	xhr	Dashboard.jsx:168	3.9 kB	
reminders	200	xhr	Dashboard.jsx:178	0.8 kB	

Backend Model Alignment with Frontend UI Requirements

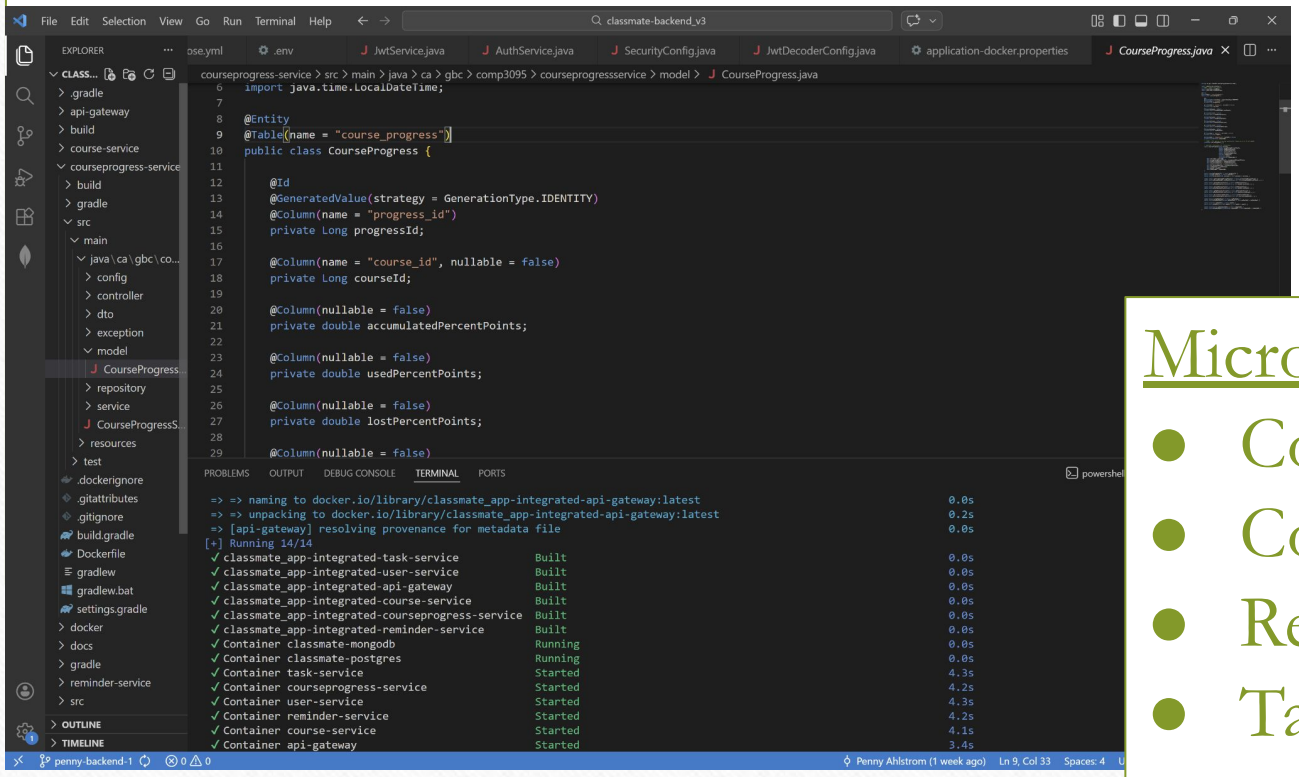
UML Class Diagram



Updates

- Grade + GradeReport -> CourseProgress
- Weekly Progress 'Snap-Shot'

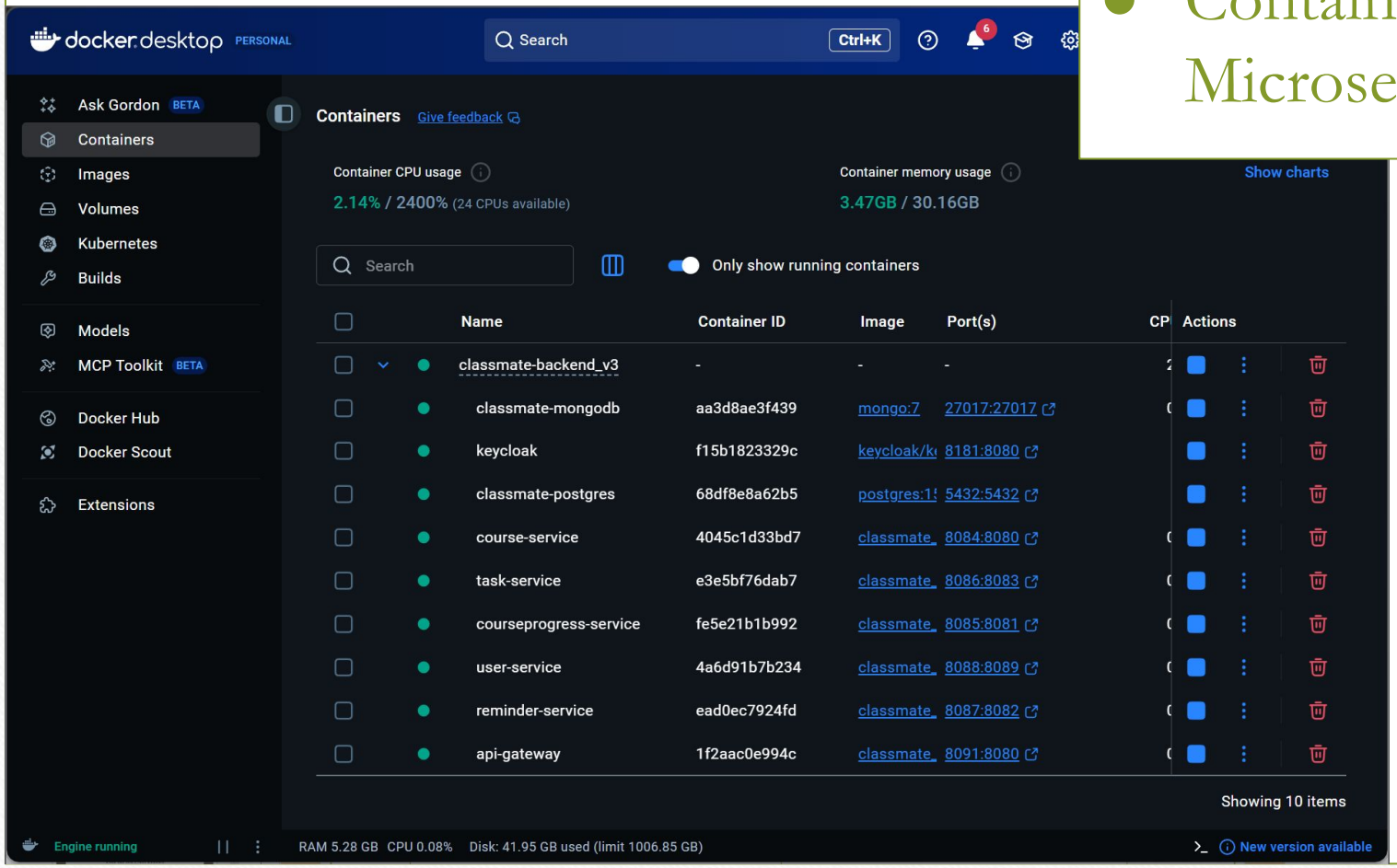
Model Updates and DB Migration Scripts



Microservices

- Course
- Course-Progress
- Reminder
- Task
- User

Containerized Solution



Docker Desktop

- Container for each Microservice

Testing Endpoints

Postman

- API Gateway
- JWT Authentication
- Status - Success

Home Workspaces API Network Search Postman Ctrl K

We're updating our plans and pricing on March 1. See our blog for more details.

COMP3095-F2025 New Import

Collections Environments History Flows Files BETA

Search collections

ClassMate-ApiGateway
- POST Register
- POST Login
- POST CreateCourse
- GET GetAllCourses
- GET GetCourseById
- PUT EditCourseById
- DEL DeleteCourseById
- POST CreateCourseProgress
- GET GetCourseProgressById
- PUT EditCourseProgressById
- DEL DeleteCourseProgressById
- GET GetAllCourseProgress
- POST CreateTask
- GET GetAllTasks
- GET GetTaskById
- PUT EditTaskById
- DEL DeleteTaskById
- POST CreateReminder
- GET GetAllReminders
- GET GetReminderById
- PUT EditReminderById
- DEL DeleteReminderById

ClassMate-DirectPort
comp3023
COMP3123 Assignment1

Overv | GET Get A | POST Regi | POST Logi | POST Crea | PUT EditTa | GET GetAll | GET GetC | POST Crea

ClassMate-ApiGateway / CreateCourse

POST http://localhost:8091/api/v1/courses Send

Docs Params Authorization Headers (9) Body Scripts Settings Cookies

Auth Type: Bearer Token

Token: ••••••••••••••••••••••••••••••••

The authorization header will be automatically generated when you send the request. Learn more about Bearer Token authorization.

Body Cookies Headers (13) Test Results — 201 Created · 26 ms · 660 B · Save Response

JSON Preview Visualize


```
{
    "courseId": 4,
    "code": "COMP3095",
    "title": "Microservices",
    "instructor": "Smith",
    "meetings": [
        {
            "dayOfWeek": 1,
            "startTime": "10:00:00",
            "endTime": "11:30:00"
        },
        {
            "dayOfWeek": 3,
            "startTime": "10:00:00",
```


Cloud View Find and replace Console Terminal Runner Start Proxy Cookies Vault Trash

Deliverables and Scope

- Revised System Design and Class Structure
- Aligned Backend with User Interface Requirements
- Incorporated Db Schema Migrations

Constraints, Assumptions and Dependencies

- JWT Auth - Data Integrity - Name and Format Consistency
- User Scoped - JSON Returns - BE and FE Validation
- Db Migration Tooling - Auth Service - API Gateway

Learning Plan Updates

- Figma Prototype - React Reusable Components
- RESTful APIs - Clean Integration Layer
- GraphQL - Query Aggregation

Next Steps

- Dashboard Page -> Course, Task, Reminder, Progress Page
- Styling -> UI Library
- Forms and Functionality

ClassMate

Always has your back — keeping track of deadlines, assignments, and progress, so you can stress less and achieve more.

Thank you! Any Questions?